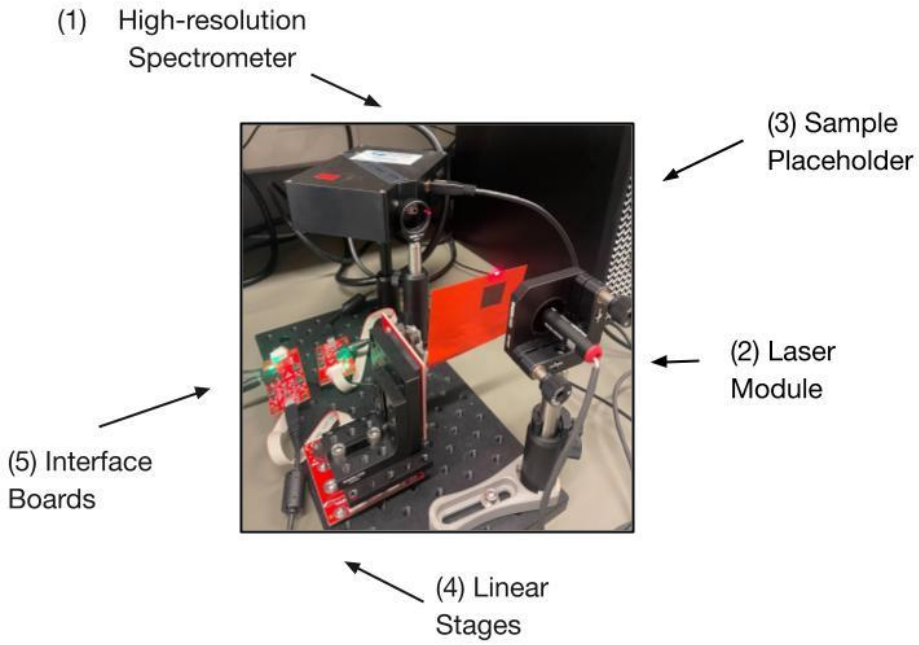
GSTEM 2025 Summer Research Internship

Overview

Hardware Setup

The experimental setup consisted of five primary components:



Photograph of the complete optical scanning system with labeled components.

Component	**Model**	**Key Specs**	**Function**
Spectrometer	Ocean Optics HR400	3,401 channels, ~200-1,100 nm, 0.1 nm resolution (nominal)	Capturing of full spectra at each position.
Laser Module	Thorlabs, 650 nm, 1 mW	Class II, continuous-wave	Illumination for reflective measurements.
Linear Stages (2x)	Thorlabs ELL17	28 mm travel, piezoelectric drive, ~100 µm repeatability	Positioning of sample in both X & Y directions.
Interface Boards	Thorlabs ELL17 Interface	USB 2.0	Communication with stages.

Sample	Adhesive paper note	—	Simulating surfaces with varying transmission properties.

Software Guide

The software is split into **four main scripts**:

1. **`linear_stage_x.py`** / **`linear_stage_y.py`**: Control classes for the two stages.
2. **`main.py`**: Stage connections, spectrometer setup, scanning logic, and saving spectra.
3. **`gui.py`**: Graphical interface for controlling scans without editing code.
4. **`plot_spectra.py`**: Loads saved spectra and generates heatmaps.

Install the following Python packages:

```
pip install elliptec seabreeze pyqt5 pandas numpy matplotlib
```

Linear Stage (X)

```
# linear_stage_x.py
# Handles communication with the X-axis Thorlabs ELL17 linear stage

import elliptec
import sys, os

class LinearStageX():
# Automatically initializes variables when the object is created
def __init__(self):
    self.ports = None         # List of available COM ports
    self.device = None        # The connected stage device object
    self._controller = None   # Controller object for sending commands


def find_devices(self):
    # Scan the computer for connected ELL17 controllers
    ports = elliptec.scan.find_ports()
    self.ports = ports[:]     # Store ports in object variable
    return ports


def connect(self, idn):
```

```
        # Create a controller for the given COM port
        self._controller = elliptec.Controller(idn, debug=True)
        self._controller.port = idn


        # Find devices connected to the controller
        self.devices = elliptec.scan_for_devices(controller=self._controller)


        # Create a Linear stage object for control
        self.device = elliptec.Linear(self._controller)
        return self.device.info  # Return device information


    def move(self, pos):
        # Move the stage to a specific position (mm)
        sys.stdout = open(os.devnull, 'w')  # Suppress stage output
        position = self.device.set_distance(pos)
        sys.stdout = sys.__stdout__
        return position
```

Linear Stage (Y)

```
# linear_stage_y.py
# Handles communication with the Y-axis Thorlabs ELL17 linear stage
# Almost identical to LinearStageX, but separated for clarity

import elliptec
import sys, os

class LinearStageY():
    def __init__(self):
        self.ports = None
        self.device = None
        self._controller = None

    def find_devices(self):
        ports = elliptec.scan.find_ports()
        self.ports = ports[:]
        return ports
```

```
    def connect(self, idn):
        self._controller = elliptec.Controller(idn, debug=True)
        self._controller.port = idn
        self.devices = elliptec.scan_for_devices(controller=self._controller)
        self.device = elliptec.Linear(self._controller)
        return self.device.info

    def move(self, pos):
        sys.stdout = open(os.devnull, 'w')
        position = self.device.set_distance(pos)
        sys.stdout = sys.__stdout__
        return position
```

Main Script

This script:

1. Connects to the two linear stages.
2. Connects to the spectrometer.
3. Performs a snake scan (alternating X direction each Y step).
4. Saves each spectrum as a CSV file.

Import and Initialize

```
# main.py
# Coordinates scanning between the X/Y linear stages and spectrometer

import seabreeze.spectrometers as sb  # Ocean Optics spectrometers
import time
import os
from PyQt5 import QtWidgets

# Import our stage classes
from linear_stage_x import LinearStageX
from linear_stage_y import LinearStageY

# Initialize stage objects
x_stage = LinearStageX()
y_stage = LinearStageY()
```

```
    # Find and connect to stages
    ports_x = x_stage.find_devices()
    ports_y = y_stage.find_devices()

    x_stage.connect('COM5')  # Replace with your X-axis COM port
    y_stage.connect('COM6')  # Replace with your Y-axis COM port

    # Initialize spectrometer
    devices = sb.list_devices()
    spectrometer = sb.Spectrometer.from_first_available()
    spectrometer.integration_time_micros(100_000)  # Exposure time in microseconds
```

Creating a Folder for Each Scan Run

```
    # Creates a new folder for each scan, incrementing run numbers
    def get_new_run_directory(base="spectra"):
        os.makedirs(base, exist_ok=True)
        existing_runs = [d for d in os.listdir(base) if d.startswith("run_") and
os.path.isdir(os.path.join(base, d))]
        run_numbers = [int(name.split("_")[1]) for name in existing_runs if
name.split("_")[1].isdigit()]
        next_run_number = max(run_numbers, default=0) + 1
        run_dir = os.path.join(base, f"run_{next_run_number}")
        os.makedirs(run_dir)
        return run_dir
```

Stop Button Component

```
    stop_requested = False

    def request_stop():
        global stop_requested
        stop_requested = True
```

Scan To (Snake Scan)

```
    # Moves the Y stage in steps, scanning X in opposite directions each row
    def scan(x_stage, y_stage, spectrometer,
             x_start=0, x_end=28, x_step=2,
             y_start=0, y_end=28, y_step=2,
             wait_time=0.1, gui=None):

        global stop_requested
        stop_requested = False

        run_dir = get_new_run_directory()

        y_positions = list(range(y_start, y_end + 1, y_step))
        for i, y_pos in enumerate(y_positions):
            if stop_requested:
                print("Scan stopped by user.")
                return

            y_stage.move(y_pos)
            print(f"Moved Y to {y_pos}")
            time.sleep(wait_time)

            if gui is not None:
                gui.y_pos.setText(str(y_pos))
                QtWidgets.QApplication.processEvents()

            # Alternate X direction each row
            x_positions = list(range(x_start, x_end + 1, x_step)) if i % 2 == 0 else
list(range(x_end, x_start - 1, -x_step))

            for x_pos in x_positions:
                if stop_requested:
                    print("Scan stopped by user.")
                    return

                x_stage.move(x_pos)
                print(f"Moved X to {x_pos}")
                time.sleep(wait_time)

                if gui is not None:
                    gui.x_pos.setText(str(x_pos))
                    QtWidgets.QApplication.processEvents()
```

```python
# Capture spectrum
wavelengths = spectrometer.wavelengths()
intensities = spectrometer.intensities()

# Save to CSV
filename = os.path.join(run_dir, f"spectrum_y{y_pos}_x{x_pos}.csv")
with open(filename, "w") as f:
    f.write("Wavelength,Intensity\n")
    for wl, inten in zip(wavelengths, intensities):
        f.write(f"{wl},{inten}\n")
print(f"Saved {filename}")
```

Design PyQt GUI

The graphical user interface for the scanning system was developed using Qt Designer and implemented in Python with the PyQt5 library. The GUI enables users to adjust key parameters, including step size, target coordinates, and scanning mode. It was intentionally designed for ease of use, ensuring that even users with minimal prior experience could quickly understand its layout and functionality.

The GUI incorporates two primary scanning modes. **“Scan To” mode** executes an automated sweep of the defined area in a continuous snake-like pattern. Users specify the “target” or endpoint coordinates and the step size, which determines the format of the scan.

Screenshot of the PyQt5 interface in full-area scan mode, with controls (left) and a schematic of the snake-pattern scan path (right).

“Single Scan” mode captures a spectrum at a specific location. Users input the target coordinates, and the stages position the spectrometer accordingly.

Screenshot of the PyQt5 interface in single-point scan mode, with coordinate entry (left) and schematic showing direct point targeting (right).

For both modes, additional control buttons include:

- **Home:** returns the stages to the origin at coordinates (0,0).
- **Stop:** immediately halts movement.
- **Go:** initiates the scan.
- **Show Spectrum:** plots the wavelength versus intensity graph for the current position.

Connect GUI and Code

```
from PyQt5.QtWidgets import QWidget
from PyQt5.uic import loadUi

# Import core objects and functions from main.py
from main import stage_x, stage_y, spec, scan, move_to_position, request_stop

# Custom QWidget subclass to control the linear stages via GUI
class StageWidget(QWidget):

    # Initialize the widget and load the UI design from a .ui file
    def __init__(self):
        super().__init__()

        loadUi('./test.ui', self)  # Loads the UI layout created in Qt Designer

        # Connect button click events to their handler methods
        self.home.clicked.connect(self.home_clicked)
        self.go.clicked.connect(self.go_clicked)
```

```
            self.stop.clicked.connect(self.stop_clicked)

        # Handler for the "Home" button: moves both stages to position (0, 0)
        def home_clicked(self):
            move_to_position(stage_x, stage_y, 0, 0)
            # Update displayed position labels
            self.x_pos.setText("0")
            self.y_pos.setText("0")

        # Handler for the "Go" button: either moves to a single point or starts a scan
        def go_clicked(self):
            # Read target positions from input fields as integers
            x_target = int(self.x_target.text())
            y_target = int(self.y_target.text())

            # Get the current selected mode from dropdown (e.g., "Single Scan" or
"Snake Scan")
            mode = self.mode_selection.currentText()

            if mode == "Single Scan":
                # Move stages once to the specified (x_target, y_target) coordinates
                move_to_position(stage_x, stage_y, x_target, y_target)
                # Update position labels accordingly
                self.x_pos.setText(str(x_target))
                self.y_pos.setText(str(y_target))
            else:
                # For scanning mode, get step sizes for X and Y directions
                x_step = int(self.x_step.text())
                y_step = int(self.y_step.text())

                # Start the scanning routine with specified parameters
                scan(stage_x, stage_y, spec,
                    x_start=0, x_end=x_target, x_step=x_step,
                    y_start=0, y_end=y_target, y_step=y_step,
                    gui=self)  # Pass self to allow GUI position updates

        # Handler for the "Stop" button: signals the scan to stop gracefully
        def stop_clicked(self):
            print("Stop button clicked")
            request_stop()

        def show_clicked(self):
```

```
        x_target = int(self.x_target.text())
        y_target = int(self.y_target.text())


        # Find the most recent run folder
        run_dirs = sorted(glob.glob("spectra/run_*"), key=os.path.getmtime)
        if not run_dirs:
            print("No run folders found.")
            return
        latest_run = run_dirs[-1]

        # Build expected file name
        file_path = os.path.join(latest_run, f"spectrum_y{y_target}_x{x_target}.csv")

        if not os.path.exists(file_path):
            print(f"No spectrum found for position ({x_target}, {y_target}).")
            return

        # Load and plot spectrum
        df = pd.read_csv(file_path)
        plt.figure(figsize=(8, 5))
        plt.plot(df["Wavelength"], df["Intensity"], label=f"({x_target}, {y_target})")
        plt.xlabel("Wavelength (nm)")
        plt.ylabel("Intensity")
        plt.title(f"Spectrum at position X={x_target}, Y={y_target}")
        plt.legend()
        plt.grid(True)
        plt.show()

from PyQt5.QtWidgets import QMainWindow

# Main application window class that holds the StageWidget
class StageGui(QMainWindow):
    def __init__(self):
        super().__init__()
        self.setWindowTitle("Stage Controller")

        # Instantiate the StageWidget and set it as the central widget
        self.stage_widget = StageWidget()
        self.setCentralWidget(self.stage_widget)

        self.show()
```

```
    # Properly handle closing the widget and then the main window
    def closeEvent(self, *args, **kwargs):
        self.stage_widget.close()
        super(QMainWindow, self).closeEvent(*args, **kwargs)


import sys
from PyQt5.QtWidgets import QApplication

# Standard Qt application entry point
if __name__ == "__main__":
    app = QApplication(sys.argv)
    window = StageGui()
    sys.exit(app.exec_())
```

Plotting Spectral Data

Once the scan is finished, we will have a folder of CSV files with names like:

`spectrum_y10_x6.csv`

Each contains two columns:

- **Wavelength** (nm)
- **Intensity** (counts)

We can load these files into Python and plot them to visualize either:

- A single point's spectrum (Show Spectrum function).
- The change in intensity across the scan (Heatmap).

```
# plot_spectra.py
# Reads all saved spectrum CSV files from a given run, extracts the intensity
# at a target wavelength, and visualizes the values as a heatmap using matplotlib.

import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Set base folder, run folder, and target wavelength for visualization
base_dir = "spectra"
```

```
    run_name = "run_20250746"
    run_path = os.path.join(base_dir, run_name)
    target_wavelength = 638  # wavelength (nm) to extract from each spectrum

    # Dictionary to store spectra for each (x, y) position
    data = {}
    for x in range(29):
        for y in range(29):
            # Build the file path for the current spectrum
            filepath = os.path.join(run_path, f"spectrum_y{y}_x{x}.csv")
            if os.path.exists(filepath):
                df = pd.read_csv(filepath)  # load spectrum into DataFrame
                data[(x, y)] = df           # store spectrum with its coordinates

    # Matrix to store extracted intensity values for each (y, x) position
    intensity_map = np.zeros((29, 29))

    # Extract intensity at the closest wavelength to the target for each spectrum
    for (x, y), df in data.items():
        idx = (df["Wavelength"] - target_wavelength).abs().idxmin()  # find closest
wavelength index
        intensity_map[y, x] = df.loc[idx, "Intensity"]               # assign intensity
to matrix

    # Plot heatmap (flipped left-right for display purposes)
    plt.imshow(np.fliplr(intensity_map), cmap='hot', origin='lower')

    plt.title(f"Heatmap at {target_wavelength} nm")
    plt.colorbar(label="Intensity")
    plt.xlabel("X Position")
    plt.ylabel("Y Position")
    plt.show()
```

How to Run the Project

Follow these steps to run the full workflow:

1. **Connect hardware**
    - Plug in the spectrometer and verify it is detected by **`seabreeze`**.
    - Connect both linear stages via USB and confirm their COM ports.

2. **Run the GUI**
   `python gui.py`

3. **View results**
   Use `plot_spectra.py` to visualize the saved spectra.

Key Notes

- The **linear stage control logic** here is based on the "Linear Stage" section, but extended for two-axis scanning.
- The **snake scan** method reduces total travel time by reversing the X direction every Y step instead of resetting.
- For best performance, adjust:
  - **Integration time** of the spectrometer (`spectrometer.integration_time_micros()`).
  - **Step size** of the stages (`x_step`, `y_step`).
  - **Wait time** after stage movement to allow vibration settling.
- Full code available at: https://github.com/lyknj/spectrometer-stage-control

About Me

Hello! My name is Anna Luo, and I'm a high school senior at Canyon Crest Academy in San Diego, California.

I had the privilege of working in the photonics lab at the ASRC under Professor Viktoriia Rutckaia as part of the NYU GSTEM summer program. Now, I hope this tutorial will help others build and use similar scanning setups for their own research and projects.